

Week 6 Progress Report

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Course: Senior Capstone – Cybersecurity

Week: Week 6

Focus: Detection Rules and Alert Generation

Overview

During Week 6, the project advanced from data storage to implementing detection capabilities. Building on the normalized event pipeline and database integration from previous weeks, rule-based detection logic was developed to identify suspicious activity within the dataset.

The system now analyzes stored events, applies detection rules, and generates structured alerts. These alerts are stored in the database and can be queried for further analysis, representing a key component of real-world SIEM functionality.

Objectives

Develop rule-based detection mechanisms, analyze stored events for suspicious patterns, generate structured alerts based on defined thresholds, store alerts in the database, and validate detection results.

Implementation Details

Detection Architecture

A modular detection layer was introduced, consisting of:

- `detect/rules.py` for individual detection rules
- `detect/engine.py` for executing all rules
- `detect_main.py` for orchestrating detection and alert generation

This structure separates detection logic from ingestion and storage, improving scalability and maintainability.

Detection Rules Implemented

The following rule-based detections were implemented:

- **SSH Invalid User Burst**
Detects multiple failed SSH login attempts from the same source IP
- **SSH Disconnect Burst**
Identifies repeated SSH connection attempts that terminate unexpectedly
- **Suspicious Honeypot Activity**
Flags honeypot interactions containing suspicious or exploit-related behavior
- **High Volume Honeypot Source**
Detects excessive activity from a single source interacting with the honeypot

These rules are threshold-based and designed to simulate common real-world attack patterns such as brute force attempts and reconnaissance activity.

Alert Generation

Each detection rule generates structured alerts containing:

- alert_id
- rule_name
- severity
- timestamp
- source
- src_ip
- description
- related_event_ids

Alerts are stored in the alerts table within the SQLite database and also written to a JSONL file for validation.

Pipeline Integration

The updated SIEM pipeline now includes detection:

Raw Logs → Ingestion → Normalization → Database Storage → Detection → Alerts

This completes a full end-to-end SIEM workflow from raw data to actionable security insights.

Results

The detection engine processed all stored events and generated alerts based on defined rules.

Detection Output:

```
=== Mini SIEM Week 6: Detection Rules ===  
Total events loaded: 1081531  
Total alerts generated: 2644  
  
Alerts by rule:  
  ssh_invalid_user_burst: 352  
  ssh_disconnect_burst: 269  
  suspicious_honeypot_activity: 1702  
  high_volume_honeypot_source: 321
```

Summary of Results:

- Total events analyzed: 1,081,531
- Total alerts generated: 2,644

Alerts by Rule:

- ssh_invalid_user_burst — 352
- ssh_disconnect_burst — 269
- suspicious_honeypot_activity — 1,702
- high_volume_honeypot_source — 321

Database Verification

The following query was used to verify alert storage:

```
SELECT COUNT(*) FROM alerts;
```

Result:

Total alerts in DB: 2644

Sample Alerts

Example alerts generated by the system:

- Detected 107 invalid SSH user attempts from 218.26.11.118
- Detected 269 invalid SSH user attempts from 67.205.20.23
- Detected 303 invalid SSH user attempts from 173.192.158.3

These results demonstrate the system's ability to identify and summarize suspicious behavior across large datasets.

Challenges Encountered

Handling large-scale data required efficient processing strategies to avoid performance issues. Additionally, designing meaningful detection thresholds required balancing sensitivity and noise to ensure alerts were both relevant and manageable.

Key Takeaways

Rule-based detection enables the transformation of raw log data into actionable insights. Structured event storage is essential for enabling detection logic, and modular design allows detection rules to be easily expanded in future iterations.

The system now closely resembles the core functionality of a SIEM platform, including ingestion, normalization, storage, detection, and alerting.

Limitations

The current detection rules are threshold-based and do not incorporate time-based correlation or advanced behavioral analysis. Additionally, due to the use of non-correlated datasets, cross-source attribution remains limited.

Future improvements will include more advanced correlation techniques and improved dataset alignment.

Next Steps (Week 7 and Beyond)

Future development will focus on improving detection capabilities and system usability. Planned enhancements include:

- Implementing time-based correlation rules
- Reducing false positives through threshold tuning
- Developing dashboards or visualization tools
- Enhancing alert analysis and reporting